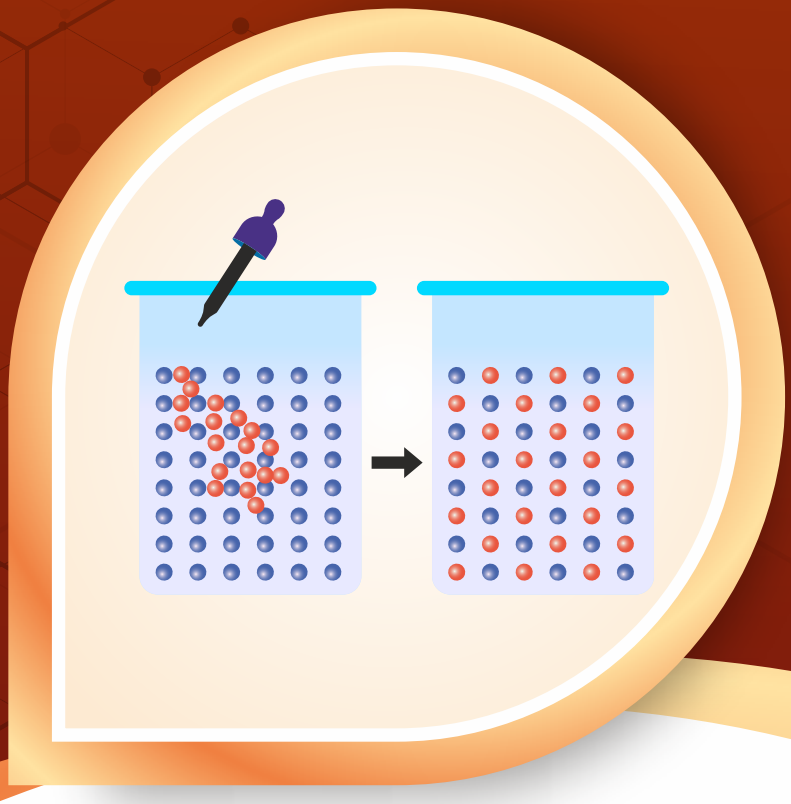
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success

PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Chemical Equilibrium

(English Medium)

EXERCISE # S-I

1. Which of the following reactions goes almost all the way to completion, and which proceeds hardly at all ?

(a) $N_2(g) + O_2(g) \rightleftharpoons 2NO\ (g)$; $K_c = 2.7 \times 10^{-18}$

(b) $2NO(g) + O_2(g) \rightleftharpoons 2NO_2\ (g)$; $K_c = 6.0 \times 10^{13}$

CE0001

2. Write down the expression for equilibrium constant K_C and K_P for the following reactions :

(a) $SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$

(b) $H_2O(\ell) \rightleftharpoons H_2O(g)$

(c) $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

(d) $CH_3COOH(\ell) + C_2H_5OH(\ell) \rightleftharpoons CH_3COOC_2H_5(\ell) + H_2O(\ell)$

(e) $NH_3(aq) + H_2O(\ell) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$
(excess)

(f) $Zn(s) + 2H^+(aq) \rightleftharpoons Zn^{2+}(aq) + H_2(g)$ **CE0002**

3. Write the relationship between K_p and K_c for the following reactions :

(a) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ (b) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(c) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ (d) $2H_2O(g) \rightleftharpoons 2H_2(g) + O_2(g)$

(e) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

CE0003

4. Which reactant is more stable from the given equilibrium reaction :

$2XO\ (g) \rightleftharpoons X_2(g) + O_2(g)$; $K_1 = 1 \times 10^{24}$

$2XO_2\ (g) \rightleftharpoons X_2(g) + 2O_2(g)$; $K_2 = 2.5 \times 10^{10}$

CE0004

5. The value of K_c for the reaction $2A \rightleftharpoons B + C$ is 2×10^{-3}. At a given time, the composition of reaction mixture is $[A] = [B] = [C] = 3 \times 10^{-4}$ M. In which direction the reaction will proceed?

CE0005

6. The ester or ethyl acetate is formed by the reaction between ethanol and acetic acid and equilibrium is represented as :

$CH_3COOH_{(l)} + C_2H_5OH_{(l)} \rightleftharpoons CH_3COOC_2H_{5(\ell)} + H_2O_{(l)}$.

(a) Write the concentration ratio (reaction quotient), Q_C for this reaction. (Note that water is not in excess and is not a solvent in this reaction).

(b) At 293 K, if starts with 1.00 mole of acetic acid and 0.180 moles of ethanol, there is 0.17 moles of ethyl acetate in the final equilibrium mixture. Calculate the equilibrium constant.

(c) Starting with 0.500 moles of ethanol and 1.000 mole of acetic acid and maintaining it at 293 K, 0.214 moles of ethyl acetate is found after some time. Has equilibrium been reached ?

CE0006

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7. If $K_c = 5.0 \times 10^{-10}$ at 1000 K for the reaction N_2 (g) + O_2 (g) $\rightleftharpoons$ 2NO (g), what is K_c (in the order of 10^9) at 1000 K for the reaction 2NO (g) $\rightleftharpoons$ N_2 (g) + O_2 (g)?

CE0007

8. An equilibrium mixture of PCl_5, PCl_3 and Cl_2 at a certain temperature contains 5.0×10^{-4} M PCl_5, 1.5×10^{-2} M PCl_3, and 3×10^{-2} M Cl_2. Calculate the equilibrium constant K_c for the reaction PCl_5 (g) $\rightleftharpoons$ PCl_3 (g) + Cl_2 (g).

CE0008

9. A sample of HI (9.6×10^{-3} mol) was placed in an empty 2.00 L container at 1000 K. After equilibrium was reached, the concentration of I_2 was 4×10^{-4} M. Calculate the value of K_c at 1000 K for the reaction H_2 (g) + I_2 (g) $\rightleftharpoons$ 2HI (g).

CE0009

10. The vapour pressure of water at 27°C is 0.24 atm. Calculate the values of K_p (in atm) and K_c (in M) at 27°C for the equilibrium H_2O (l) $\rightleftharpoons$ H_2O (g). (R = 0.08 L-atm/K-mol)

CE0010

11. The value of K_c for the reaction $3O_2$ (g) $\rightleftharpoons$ $2O_3$ (g) is 8×10^{-55} at 25°C. Do you expect pure air at 25°C to contain much O_3 (ozone) when O_2 and O_3 are in equilibrium? If the equilibrium concentration of O_2 in air at 25°C is 8×10^{-3} M, what is the equilibrium concentration of O_3?

CE0011

12. One mole of H_2O and one mole of CO are taken in a 10 L vessel and heated to 725 K. At equilibrium 40 percent of water (by mass) reacts with carbon monoxide according to the equation.

$$H_2O_{(g)} + CO_{(g)} \rightleftharpoons H_{2(g)} + CO_{2(g)}$$

Calculate the equilibrium constant for the reaction.

CE0012

13. A vessel at 1000 K contains CO_2 with a pressure of 0.6 atm. Some of the CO_2 is converted into CO on addition of graphite. Calculate the value of K_P (in atm), if total pressure at equilibrium is 0.9 atm.

CE0013

14. Calculate the equilibrium concentration of HI (in M) at 300 K if initially 2 mol each of H_2 and I_2 are taken in a closed container of having volume 10 lit.

[Given : $H_2 + I_2 \rightleftharpoons 2HI$; K = 100 at 300 K]

CE0014

15. A mixture of SO_3, SO_2 and O_2 gases is maintained at equilibrium in 10 litre flask at a particular temperature in which K_c for the reaction,

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \text{ is } 100 \text{ mol}^{-1} \text{ litre.}$$

At equilibrium,

(a) If number of mole of SO_3 and SO_2 in flask are same, how many mole of O_2 are present ?

(b) If number of mole of SO_3 in flask are twice the number of mole of SO_2, how many mole of O_2 are present ?

CE0015

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16. For the given reaction,

$A(g) + B(g) \rightleftharpoons C(g) + D(g)$

the number of moles at equilibrium was found to be 4, 4 , 2 & 8 respectively in 1 L container. Find the new equilibrium concentration of C (in M) if 18 moles of D are added to above 1 L container.

CE0016

17. For the given reaction,

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

the number of moles at equilibrium was found to be 2, 6 & 4 respectively at 10 atm pressure. Find the new equilibrium moles of PCl_5, if it is restablished at an equilibrium pressure 50 atm.

CE0017

18. For the dissociation reaction $N_2O_4\ (g) \rightleftharpoons 2NO_2\ (g)$ derive the expression for the degree of dissociation in terms of K_p and total pressure P.

CE0018

19. For the reaction $2SO_{3(g)} \rightleftharpoons 2SO_{2(g)} + O_{2(g)}$

Initially 5 moles of SO_3 is taken and at equillibrium 4 moles of SO_3 is dissociated, find out the value of K_C (in M) (Volume = 2litre)

CE0019

20. 1.1 mole of A mixed with 2.2 mole of B and the mixture is kept in a 1 litre flask at the equillibrium, $A + 2B \rightleftharpoons 2C + D$ is reached. At equillibrium 0.2 mole of C is formed then the value of $(2000 \times K_C)$ will be.

CE0020

21. The air pollutant NO is produced in automobile engines from the high temperature reaction $N_2(g) + O_2(g) \rightleftharpoons 2NO\ (g)$; $K_c = 16$ at 2300 K. If the initial concentrations of N_2 and O_2 at 2300 K are both 1.5 M, what are the concentrations of N_2 (in M) when the reaction mixture reaches equilibrium?

CE0021

22. At 700 K, $K_p = 0.2$ atm for the reaction $ClF_3\ (g) \rightleftharpoons ClF\ (g) + F_2(g)$. Calculate the equilibrium partial pressure of ClF_3, if only ClF_3 is present initially, at a partial pressure of 0.150 atm.

CE0022

23. The degree of dissociation of N_2O_4 into NO_2 at 1.5 atmosphere and 40°C is 0.25. Calculate its degree of dissociation at 10 atmospheric pressure at same temperature.

CE0023

24. At 46°C, K_p for the reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ is 0.667 atm . Compute the percent dissociation of N_2O_4 at 46° C at a total pressure of 380 Torr .

CE0024

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25. 2 NOBr (g) $\rightleftharpoons$ 2 NO (g) + Br_2 (g). If nitrosyl bromide (NOBr) is (100/3)% dissociated at 25° C and a total pressure of 0.28 atm. Calculate K_p for the dissociation at this temperature.

CE0025

26. The system $N_2O_4 \rightleftharpoons 2NO_2$ maintained in a closed vessel at 60°C & a pressure of 5 atm has an average (i.e. observed) molecular weight of 69. At what pressure (in atm) at the same temperature would the observed molecular weight be (230/3) ?

CE0026

27. In the esterfication C_2H_5OH (l) + CH_3COOH (l) $\rightleftharpoons$ $CH_3COOC_2H_5$ (l) + H_2O (l) an equimolar mixture of alcohol and acid taken initially yields under equilibrium, the water with mole fraction = 0.333. Calculate the equilibrium constant.

CE0027

28. The equilibrium constant for the reaction is 9 at 900°C $S_2(g) + C(s) \rightleftharpoons CS_2(g)$. Calculate the pressure (in atm) of $CS_2(g)$ at equilibrium, when 2 atm of S_2 and excess of C(s) come to equilibrium.

CE0028

29. A sample of $CaCO_3(s)$ is introduced into a sealed container of volume 0.82 litre & heated to 1000K until equilibrium is reached. The equilibrium constant for the reaction $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ is 4×10^{-2} atm at this temperature. Calculate the mass (in mg) of CaO present at equilibrium.

CE0029

30. Anhydrous calcium chloride is often used as a dessicant. In the presence of excess of $CaCl_2$, the amount of the water taken up is governed by $K_p = 6.4 \times 10^{85}$ (atm^{-6}) for the following reaction at room temperature, $CaCl_2(s) + 6H_2O(g) \rightleftharpoons CaCl_2 . 6H_2O(s)$. What is the equilibrium partial pressure (in 10^{-15} atm) of water vapour in a closed vessel that contains $CaCl_2(s)$?

CE0030

31. 20.0 grams of $CaCO_3(s)$ were placed in a closed vessel, heated & maintained at 727° C under equilibrium $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ and it is found that 75 % of $CaCO_3$ was decomposed. What is the value of K_p (in atm) ? The volume of the container was 15 litres.

CE0031

32. **(a)** Methanol, a liquid fuel that could possibly replace gasoline, can be prepared from water gas and additional hydrogen at high temperature and pressure in the presence of a suitable catalyst. Write the expression for the equilibrium constant for the reversible reaction.

$2H_2$ (g) + CO (g) $\rightleftharpoons$ CH_3OH (g) $\quad \Delta H = -90.2$ kJ

(b) Assume that equilibrium has been established and predict how the concentration of H_2, CO and CH_3OH will differ at a new equilibrium if (1) more H_2 is added. (2) CO is removed. (3) CH_3OH is added. (4) the pressure on the system is increased. (5) the temperature of the system is increased. (6) more catalyst is added.

CE0032

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33. Ammonia is a weak base that reacts with water according to the equation

$NH_3\ (aq) + H_2O\ (l) \rightleftharpoons NH_4^+ + OH^-\ (aq)$

Will any of the following increase the percent of ammonia that is converted to the ammonium ion in water?

(a) Addition of NaOH. (b) Addition of HCl. (c) Addition of NH_4Cl.

CE0033

34. Consider the gas-phase hydration of hexafluoroacetone, $(CF_3)_2CO$:

$$(CF_3)_2CO\ (g) + H_2O\ (g) \underset{k_r}{\overset{k_f}{\rightleftharpoons}} (CF_3)_2C(OH)_2\ (g)$$

At 76°C, the forward and reverse rate constants are $k_f = 0.15\ M^{-1}s^{-1}$ and $k_r = 6 \times 10^{-4}s^{-1}$. What is the value of the equilibrium constant K_c (in M^{-1}) ?

CE0034

35. Listed in the table are forward and reverse rate constants for the reaction $2NO\ (g) \rightleftharpoons N_2(g) + O_2(g)$

Temperature (K)	$\mathbf{k_f\ (M^{-1}s^{-1})}$	$\mathbf{k_r\ (M^{-1}s^{-1})}$
1400	0.29	1.1×10^{-6}
1500	1.3	1.4×10^{-5}

Is the reaction endothermic or exothermic? Explain in terms of kinetics.

CE0035

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EXERCISE # S-II

1. One mole of N_2O_4 (g) at 300 K is left in a closed container under one atm . It is heated to 600 K when 20 % by mass of N_2O_4 (g) decomposes to NO_2 (g) . Calculate resultant pressure (in atm).

CE0036

2. Solid Ammonium carbamate dissociates as: $NH_2\ COONH_4$ (s) $\rightleftharpoons$ $2NH_3(g) + CO_2(g)$. In a closed vessel solid ammonium carbamate is in equilibrium with its dissociation products. At equilibrium, ammonia is added such that the partial pressure of NH_3 at new equilibrium now equals the original total pressure. The ratio of total pressure at new equilibrium to that of original total pressure is x : 3, the value of 'x' is.

CE0037

3. A definite amount of solid NH_4HS is placed in a flask already containing ammonia gas at a certain temperature and 0.50 atm pressure. NH_4HS decomposes to give NH_3 and H_2S and at equilibrium total pressure in flask is 0.84 atm. Calculate equilibrium constant for the reaction (in atm^2) :

CE0038

4. For the reaction $A(g) + 2B(g) \rightleftharpoons C(g) + D(g)$; $K_c = 10^{12}$.
If the initial moles of A,B,C and D are 0.5, 1, 0.5 and 3.5 moles respectively in a one litre vessel. What is the equilibrium concentration of B (in millimole/litre) ?

CE0039

5. The ratio of equilibrium molar concentration of PCl_3 and PCl_5 at 300 K if initially 1 mol of PCl_5 is taken in a closed container of having volume 10 litre, is x : 1.00. The value of 'x' is.
[Given : $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$; $K_C = 10^{-2}$ at 300 K, $\sqrt{41} = 6.4$]

CE0040

6. When 36.8g N_2O_4 (g) is introduced into a 1.0-litre flask at 27°C . The following equilibrium reaction occurs :
N_2O_4 (g) $\rightleftharpoons$ $2NO_2$ (g) ; $K_p = 0.164$ atm.
(a) Calculate K_c of the equilibrium reaction. (in 10^{-3} M)
(b) What are the number of millimoles of N_2O_4 at equilibrium?
(c) What is the total pressure of gases (in atm) in the flask at equilibrium?
(d) What is the percent dissociation of N_2O_4?

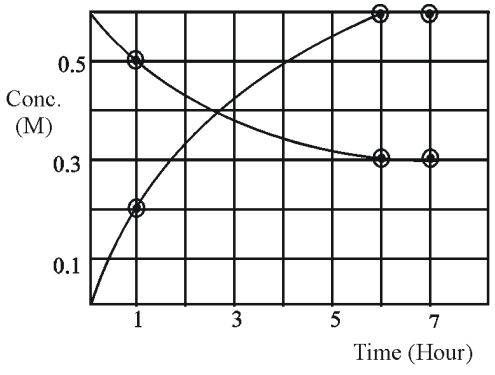
CE0041

7. The progress of the reaction

$A \rightleftharpoons nB$ with time, is presented in figure. Determine

(i) the value of n.

(ii) the equilibrium constant, K (in M^{n-1}).



CE0042

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8. A mixture of hydrogen & iodine in the mole ratio 1.5 : 1 is maintained at 450° C. After the attainment of equilibrium $H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$, it is found on analysis that the mole ratio of I_2 to HI is 1 : 18. Calculate the equilibrium constant, if initially, 127 grams of iodine were taken. (I = 127)

CE0043

9. At certain temperature (T) for the gas phase reaction

$2H_2O(g) + 2Cl_2(g) \rightleftharpoons 4HCl(g) + O_2(g)$ $\quad K_p = 12 \times 10^8$ atm

If Cl_2, HCl & O_2 are mixed in such a manner that the partial pressure of each is 2 atm and the mixture is brough into contact with excess of liquid water. What would be approximate partial pressure (in 10^{-3} atm) of Cl_2 when equilibrium is attained at temperature (T)?

[Given : Vapour pressure of water is 380 mm Hg at temperature (T)]

CE0044

10. Two solids X and Y disssociate into gaseous products at a certain temperature as follows:

$X(s) \rightleftharpoons A(g) + C(g)$, and $Y(s) \rightleftharpoons B(g) + C(g)$. At a given temperature, pressure over excess solid X is 40 mm and total pressure over solid Y is 60 mm. When they are preset in separate containers. Calculate

(a) the values of K_p for two reactions (in mm^2)

(b) the ratio of moles of A and B in the vapour state over a mixture of X and Y.

(c) the total pressure of gases (in mm) over a mixture of X and Y.

CE0045

11. When 1 mole of A(g) is introduced in a closed rigid 1 litre vessel maintained at constant temperature the following equilibria are established.

$A(g) \rightleftharpoons B(g) + C(g) : K_{C_1}$

$C(g) \rightleftharpoons D(g) + B(g) : K_{C_2}$

The pressure at equilibrium is twice the initial pressure. Calculate the value of $\frac{K_{C_2}}{K_{C_1}}$ if $\frac{[C]_{eq}}{[B]_{eq}} = \frac{1}{5}$

CE0046

12. Equilibrium constants are given (in atm) for the following reactions at 0° C:

$SrCl_2 \cdot 6H_2O(s) \rightleftharpoons SrCl_2 \cdot 2H_2O(s) + 4H_2O(g)$ $\quad K_p = 5 \times 10^{-12}$

$Na_2HPO_4 \cdot 12\,H_2O(s) \rightleftharpoons Na_2HPO_4 \cdot 7\,H_2O(s) + 5H_2O(g)$ $\quad K_p = 2.43 \times 10^{-13}$

$Na_2SO_4 \cdot 10\,H_2O(s) \rightleftharpoons Na_2SO_4(s) + 10\,H_2O(g)$ $\quad K_p = 1.024 \times 10^{-27}$

$$\left(\%\ \text{Relative humidity} = \frac{P_{H_2O}}{\text{Vapour pressure of } H_2O} \times 100\right)$$

The vapor pressure of water at 0°C is 4.56 torr.

Out of $SrCl_2 \cdot 2H_2O(s)$, $Na_2HPO_4 \cdot 7\,H_2O(s)$ and $Na_2SO_4(s)$,

Which is the most effective drying agent at 0°C?

CE0047

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13. At certain temperature , K_C = 1.0 for reaction :

$$NO_2(g) + NO\ (g) \rightleftharpoons N_2O(g) + O_2(g)$$

equal moles of NO and NO_2 are to be placed in 5 litre container until N_2O concentration at equilibrium is 0.5M. How many mole of (NO + NO_2) must be placed in the container.

CE0048

14. In a vessel, two equilibrium are simultaneously established at same temperature as follows,

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad .....(i)$$

$$N_2(g) + 2H_2(g) \rightleftharpoons N_2H_4(g) \quad .....(ii)$$

Initially the vessel contains N_2 and H_2 in the molar ratio of 9 : 13. The equilibrium pressure is 7 P_0 in which pressure due to ammonia is P_0 and due to hydrogen is $2P_0$. Find the values of equilibrium constants (K_p's) for both the reactions.

CE0049

15. If vapour density of equilibrium mixture NO_2 and N_2O_4 is found to be 40 for the reaction,

$N_2O_4 \rightleftharpoons 2NO_2$. Find :

(a) Molar mass of equilibrium mixture (in gm/mol)

(b) Degree of dissociation

(c) Percentage of NO_2 in the mixture

CE0050

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EXERCISE # O-I

ONLY ONE IS CORRECT :

1. x $\rightleftharpoons$ y reaction is said to be in equilibrium, when :-

(A) Only 10% conversion of x to y takes place
(B) Complete conversion of x to y has taken place
(C) Conversion of x to y is only 50% complete
(D) The rate of change of x to y is just equal to the rate of change of y to x in the system

CE0051

2. For which reaction is $K_p = K_c$:-

(A) $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$ (B) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(C) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ (D) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

CE0052

3. For the reaction

$CuSO_4.5H_2O(s) \rightleftharpoons CuSO_4.3H_2O_{(s)} + 2H_2O_{(g)}$

Which one is correct representation :–

(A) $K_p = (P_{(H_2O)})^2$ (B) $K_c = [H_2O]^2$ (C) $K_p = K_c(RT)^2$ (D) All

CE0053

4. $\log \frac{K_p}{K_c} + \log RT = 0$ is true relationship for the following reaction:-

(A) $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ (B) $2SO_2 + O_2 \rightleftharpoons 2SO_3$

(C) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (D) (B) and (C) both

CE0054

5. For a reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$, the value of K_c does not depends upon :-

(a) Initial concentration of the reactants (b) Pressure
(c) Temperature (d) Catalyst

(A) Only c (B) a, b, c (C) a, b, d (D) a, b, c, d

CE0055

6. If some He gas is introduced into the equilibrium $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ at constant pressure and temperature then equilibrium constant of reaction:

(A) Increase (B) Decrease (C) Unchange (D) Nothing can be said

CE0056

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7. The equilibrium constant for the given reaction :

$$SO_3(g) \rightleftharpoons SO_2(g) + \frac{1}{2}O_2(g) \; ; \; K_c = 5 \times 10^{-2}$$

The value of K_c for the reaction :
$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, will be
(A) 400 (B) 2.40×10^{-3} (C) 9.8×10^{-2} (D) 4.9×10^{-2}

CE0057

8. For the following three reactions, 1, 2 and 3 equilibrium, constants are given :

(1) $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$; K_1

(2) $CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$; K_2

(3) $CH_4(g) + 2H_2O(g) \rightleftharpoons CO_2(g) + 4H_2(g)$; K_3

Which of the following relations is correct ?

(A) $K_1\sqrt{K_2} = K_3$ (B) $K_2K_3 = K_1$ (C) $K_3 = K_1K_2$ (D) $K_3 = K_2^3K_1^2$

CE0058

9. Sulfide ion in alkaline solution reacts with solid sulfur to form polysulfide ions having formulas S_2^{2-}, S_3^{2-}, S_4^{2-} and so on. The equilibrium constant for the formation of S_2^{2-} is 12 (K_1) & for the formation of S_3^{2-} is 132 (K_2), both from S and S^{2-}.What is the equilibrium constant for the formation of S_3^{2-} from S_2^{2-} and S?
(A) 11 (B) 12 (C) 132 (D) None of these

CE0059

10. If 0.5 mole H_2 is reacted with 0.5 mole I_2 in a ten-litre container at 444° C and at same temperature value of equilibrium constant K_c is 49, the ratio of [HI] and [I_2] will be :-

(A) 7 (B) $\frac{1}{7}$ (C) $\sqrt{\frac{1}{7}}$ (D) 49

CE0060

11. In the reaction, $N_2 + O_2 \rightleftharpoons 2NO$, the moles/litre of N_2, O_2 and NO respectively 0.25, 0.05 and 1.0 at equilibrium. The initial concentrations of N_2 and O_2 will respectively be
(A) 0.75 mole/lit., 0.55 mole/lit. (B) 0.50 mole/lit., 0.75 mole/lit.
(C) 0.25 mole/lit., 0.50 mole/lit (D) 0.25 mole/lit., 1.0 mole/lit.

CE0061

12. How many moles per litre of PCl_5 has to be taken to obtain 0.1 mole Cl_2, if the value of equilibrium constant K_c is 0.04 ?
(A) 0.15 (B) 0.25 (C) 0.35 (D) 0.05

CE0062

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\4-Ideal-Real-Atomic-Chemical Eq\Eng\4. Che. Eq\02. Ex. p65

13. In the beginning of the reaction, $A \rightleftharpoons B + C$, 2 moles of A are taken, out of which 0.5 mole gets dissociated. What is the degree of dissociation of A ?

(A) 0.5 (B) 1 (C) 0.25 (D) 4.2

CE0063

14. In the reaction, $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, the amount of each PCl_5, PCl_3 and Cl_2 is 2 mole at equilibrium and total pressure is 3 atmosphere. The value of K_p will be

(A) 1.0 atm. (B) 3.0 atm. (C) 2.9 atm. (D) 6.0 atm.

CE0064

15. 4 moles of PCl_5 are heated at constant temperature in closed container. If degree of dissociation for PCl_5 is 0.5 calculate total number of moles at equilibrium :-

(A) 4.5 (B) 6 (C) 3 (D) 4

CE0065

16. For the reaction $A + 2B \rightleftharpoons 2C + D$, initial concentration of A is a and that of B is 1.5 times that of A. Concentration of A and D are same at equilibrium. What should be the concentration of B at equilibrium ?

(A) $\frac{a}{4}$ (B) $\frac{a}{2}$ (C) $\frac{3a}{4}$ (D) All of the above.

CE0066

17. For the reaction $3\,A(g) + B(g) \rightleftharpoons 2\,C(g)$ at a given temperature, $K_c = 9.0$. What must be the volume of the flask, if a mixture of 2.0 mol each of A, B and C exist in equilibrium?

(A) 6L (B) 9L (C) 36 L (D) None of these

CE0067

18. For the following gases equilibrium. $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

K_p is found to be equal to K_c. This is attained when temperature is

(A) 0°C (B) 273 K (C) 1 K (D) 12.19 K

CE0068

19. The degree of dissociation of SO_3 is α at equilibrium pressure p^0.

K_p for $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$

(A) $\frac{p^0\alpha^3}{2(1-\alpha)^3}$ (B) $\frac{p^0\alpha^3}{(2+\alpha)(1-\alpha)^2}$ (C) $\frac{p^0\alpha^2}{2(1-\alpha)^2}$ (D) None of these

CE0069

20. For the reaction : $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$, the degree of dissociated (α) of HI(g) is related to equilibrium constant K_P by the expression

(A) $\frac{1+2\sqrt{K_p}}{2}$ (B) $\sqrt{\frac{1+2K_p}{2}}$ (C) $\sqrt{\frac{2K_p}{1+2K_p}}$ (D) $\frac{2\sqrt{K_p}}{1+2\sqrt{K_p}}$

CE0070

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\04-Modi-Ross-Atomic-Chemical Eq\Eng\4.Che. Eq\02. Ex..p65

21. The equilibrium constant for the reaction

$A(g) + 2B(g) \rightleftharpoons C(g)$

is 0.25 dm^6mol^{-2}. In a volume of 5 dm^3, what amount of A must be mixed with 4 mol of B to yield 1 mol of C at equilibrium.

(A) 3 moles (B) 24 moles (C) 26 moles (D) None of these

CE0071

22. A 20.0 litre vessel initially contains 0.50 mole each of H_2 and I_2 gases. These substances react and finally reach an equilibrium condition. Calculate the equilibrium concentration of HI if $K_{eq} = 49$ for the reaction $H_2 + I_2 \rightleftharpoons 2HI$.

(A) 0.78 M (B) 0.039 M (C) 0.033 M (D) 0.021 M

CE0072

23. At 675 K, $H_2(g)$ and CO_2 (g) react to form CO(g) and H_2O (g), K_p for the reaction is 0.16. If a mixture of 0.25 mole of $H_2(g)$ and 0.25 mol of CO_2 is heated at 675 K, mole % of CO(g) in equilibrium mixture is :

(A) 7.14 (B) 14.28 (C) 28.57 (D) 33.33

CE0073

24. The vapour density of N_2O_4 at a certain temperature is 30. What is the % dissociation of N_2O_4 at this temperature?

(A) 53.3% (B) 106.6% (C) 26.7% (D) None

CE0074

25. The equilibrium constant K_P (in atm) for the reaction is 9 at 7 atm and 300 K.

$A_2(g) \rightleftharpoons B_2(g) + C_2(g)$

Calculate the average molar mass (in gm/mol) of an equilibrium mixture.

Given : Molar mass of A_2, B_2 and C_2 are 70, 49 & 21 gm/mol respectively.

(A) 50 (B) 45 (C) 40 (D) 37.5

CE0075

26. Vapour density of the equilibrium mixture of the reaction

$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$ is 6.0

Percent dissociation of ammonia gas is:

(A) 13.88 (B) 58.82 (C) 41.66 (D) None of these

CE0076

27. The equilibrium concentration of B $[(B)_e]$ for the reversible reaction $A \rightleftharpoons B$ can be evaluated by the expression:-

(A) $K_C[A]_e^{-1}$ (B) $\frac{k_f}{k_b}[A]_e^{-1}$ (C) $k_f k_b^{-1}[A]_e$ (D) $k_f k_b [A]^{-1}$

CE0077

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\4-Ideal-Real-Atomic-Chemical Eq\Eng\4. Che. Eq\02. Ex. p65

28. The equilibrium $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ shifts forward if :-

(A) A catalyst is used.

(B) An adsorbent is used to remove SO_3 as soon as it is formed.

(C) Small amounts of reactants are removed.

(D) None of these

CE0078

29. In manufacture of NO, the reaction $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$, ΔH +ve is favourable if :-

(A) Pressure is increased

(B) Pressure is decreased

(C) Temperature is increased

(D) Temperature is decreased

CE0079

30. In which of the following reactions, increase in the pressure at constant temperature does not affect the moles at equliibrium :

(A) $2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$

(B) $C(g) + \frac{1}{2}O_2(g) \rightleftharpoons CO(g)$

(C) $H_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons H_2O(g)$

(D) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

CE0080

31. Change in volume of the system does not alter the number of moles in which of the following equilibrium

(A) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

(B) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(C) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(D) $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$

CE0081

32. The conditions favourable for the reaction :

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$; $\Delta H^\circ = -198$ kJ

are :

(A) low temperature, high pressure

(B) any value of T and P

(C) low temperature and low pressure

(D) high temperature and high pressure

CE0082

33. Densities of diamond and graphite are 3.5 and 2.3 gm/mL.

C (diamond) $\rightleftharpoons$ C (graphite) $\quad \Delta_r H = -1.9$ kJ/mole

favourable conditions for formation of diamond are

(A) high pressure and low temperature

(B) low pressure and high temperature

(C) high pressure and high temperature

(D) low pressure and low temperature

CE0083

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\4-Ideal-Real-Atomic-Chemical Eq\Eng\4.Che. Eq\02. Ex..p65

34. The equilibrium $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$ is attained at 25°C in a closed rigid container and an inert gas, helium is introduced. Which of the following statements is/are correct.

(A) concentrations of SO_2, Cl_2 and SO_2Cl_2 do not change

(B) more chlorine is formed

(C) concentration of SO_2 is reduced

(D) more SO_2Cl_2 is formed

CE0084

35. The yield of product in the reaction

$$2A(g) + B(g) \rightleftharpoons 2C(g) + Q \text{ kJ}$$

would be lower at :

(A) low temperature and low pressure

(B) high temperature & high pressure

(C) low temperature and to high pressure

(D) high temperature & low pressure

CE0085

ALLEN®

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EXERCISE # O-II

MORE THAN ONE MAY BE CORRECT :

1. Following two equilibrium is simultaneously established in a container

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$
$$CO(g) + Cl_2(g) \rightleftharpoons COCl_2(g)$$

If some Ni(s) is introduced in the container forming Ni $(CO)_4$ (g) then at new equilibrium

(A) PCl_3 concentration will increase (B) PCl_3 concentration will decrease

(C) Cl_2 concentration will remain same (D) CO concentration will remain same **CE0086**

2. For the reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, the forward reaction at constant temperature is favoured by

(A) introducing an inert gas at constant volume (B) introducing chlorine gas at constant volume

(C) introducing an inert gas at constant pressure (D) introducing PCl_5 at constant volume.

CE0087

3. When $NaNO_3$ is heated in a closed vessel, oxygen is liberated and $NaNO_2$ is left. At equilibrium

(A) addition of $NaNO_2$ favours reverse reaction

(B) addition of $NaNO_3$ favours forward reaction

(C) increasing temperature favours forward reaction

(D) increasing pressure favours reverse reaction

CE0088

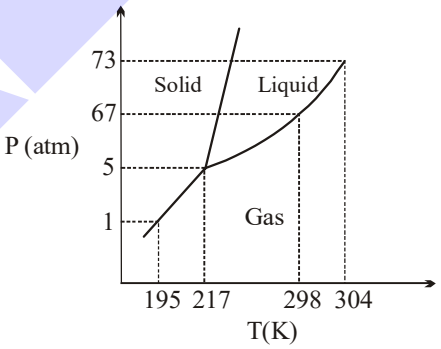
4. For the gas phase reaction, $C_2H_4 + H_2 \rightleftharpoons C_2H_6$ ($\Delta H = -32.7$ kcal), carried out in a closed vessel, the equilibrium moles of C_2H_4 can be increased by

(A) increasing the temperature (B) decreasing the pressure

(C) removing some H_2 (D) adding some C_2H_6

CE0089

5. Phase diagram of CO_2 is shown as following



Based on above find the correct statement(s)

(A) 298K is the normal boiling point of liquid CO_2

(B) At 1 atm & 190 K CO_2 will exist as gas.

(C) CO_2(s) will sublime above 195K under normal atmospheric pressure

(D) Melting point & boiling point of CO_2 will increase on increasing pressure **CE0090**

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6. The equilibrium between, gaseous isomers A, B and C can be represented as

Reaction		**Equilibrium constant**
$A(g) \rightleftharpoons B(g)$	:	$K_1 = ?$
$B(g) \rightleftharpoons C(g)$	:	$K_2 = 0.4$
$C(g) \rightleftharpoons A(g)$	:	$K_3 = 0.6$

If one mole of A is taken in a closed vessel of volume 1 litre, then

(A) [A] + [B] + [C] = 1 M at any time of the reactions

(B) Concentration of C is 4.1 M at the attainment equilibrium in all the reactions

(C) The value of K_1 is $\frac{1}{0.24}$

(D) Isomer [A] is least stable as per thermodynamics.

CE0091

7. For the gas phase exothermic reaction, $A_2 + B_2 \rightleftharpoons C_2$, carried out in a closed vessel, the equilibrium moles of A_2 can be increased by

(A) increasing the temperature (B) decreasing the pressure

(C) adding inert gas at constant pressure (D) removing some C_2

CE0092

8. Consider the equilibrium $HgO(s) + 4I^-(aq) + H_2O(l) \rightleftharpoons HgI_4^{2-}(aq) + 2OH^-(aq)$, which changes will decrease the equilibrium concentration of HgI_4^{2-}

(A) Addition of 0.1 M HI (aq) (B) Addition of HgO (s)

(C) Addition of H_2O (l) (D) Addition of KOH (aq)

CE0093

9. Decrease in the pressure for the following equilibria : $H_2O(s) \rightleftharpoons H_2O(l)$ result in the :

(A) formation of more H_2O (s) (B) formation of more $H_2O(l)$

(C) increase in melting point of H_2O(s) (D) decrease in melting point of H_2O(s)

CE0094

ASSERTION REASON :

10. **Statement -1 :** Total number of moles in a closed system at new equilibrium is less than the old equilibrium if some amount of a substance is removed from a system (consider a reaction **A(g)** $\rightleftharpoons$ **B(g)**) at equilibrium.

Statement -2 : The number of moles of the substance which is removed, is partially compensated as the system reached to new equilibrium.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is false, statement-2 is true.

(D) Statement-1 is true, statement-2 is false.

CE0095

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\4-Bksl-Rass-Atomic-Chemical Eq\Eng\4. Che. Eq\02. Ex. p65

11. **Statement-1 :** Ammonia at a pressure of 10 atm and CO_2 at a pressure of 20 atm are introduced into an evacuated chamber. If K_p for the reaction NH_2COONH_4 (s) $\rightleftharpoons$ $2NH_3$ (g) + CO_2 (g) is 2020 atm^3, the total pressure after a long time is less than 30 atm.
Statement-2 : Equilibrium can be attained from both directions.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CE0096

12. **Statement-1 :** Catalyst does not alter the equilibrium constant
Statement-2 : Because for the catalysed reaction and uncatalysed reaction ΔH remain same an equilibrium constant depend on ΔH.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CE0097

13. **Statement-1** : Equilibrium constant may show higher or lower values with increase in temperature
Statement-2 : The change depend on heat of reaction.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true. **CE0098**

COMPREHENSION :

Paragraph for Question Nos. 14 to 17

In a 7.0 L evacuated chamber, 0.50 mol H_2 and 0.50 mol I_2 react at 427°C . $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$. At the given temperature, $K_C = 49$ for the reaction.

14. What is the value of K_p ?
(A) 7 (B) 49 (C) 24.5 (D) None

CE0099

15. What is the total pressure (atm) in the chamber?
(A) 83.14 (B) 831.4 (C) 8.21 (D) None

CE0099

16. How many moles of the iodine remain unreacted at equilibrium?
(A) 0.388 (B) 0.112 (C) 0.25 (D) 0.125

CE0099

17. What is the partial pressure (atm) of HI in the equilibrium mixture?
(A) 6.385 (B) 12.77 (C) 40.768 (D) 646.58

CE0099

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Paragraph for Question Nos. 18 to 20

Influence of pressure, temperature, concentration and addition of inert gas on a reversible chemical reaction in equilibrium can be explained by formulating the expression for equilibrium constant K_c or K_p for the equilibrium. On the other hand Le Chatelier principle can be theoretically used to explain the effect of P, T or concentration on the physical or chemical equilibrium both.

18. For the reaction : $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ increase of pressure shows :

(A) An increase in degree of dissociation and a decrease in K_c
(B) A decrease in degree of dissociation and a decrease in K_c
(C) An increase in degree of dissociation but K_c remains constant
(D) A decrease in degree of dissociation but K_c remains constant

CE0100

19. For the reaction : $2SO_2 + O_2 \rightleftharpoons 2SO_3$; ΔH = –ve, An increase in temperature shows :

(A) More dissociation of SO_3 and a decrease in K_c
(B) Less dissociation of SO_3 and an increase in K_c
(C) More dissociation of SO_3 and an increase in K_c
(D) Less dissociation of SO_3 and a decrease in K_c

CE0101

20. For the reaction : $\underset{\text{Yellow}}{Fe^{3+}(aq.)} + SCN^-(aq.) \rightleftharpoons \underset{\text{Red}}{[Fe(SCN)]^{2+}(aq.)}$; in equilibrium if little more aqueous solution of $FeCl_3$ is added, than :

(A) Forward reaction becomes more and red colours is deepened
(B) Backward reaction becomes more and red colour faintens
(C) Solution becomes colourless
(D) No change will occur

CE0102

MATCH THE COLUMN :

21.

	Column I		**Column II**
(A)	$K_P < K_C$	(P)	$N_2 + 3H_2 \rightleftharpoons 2NH_3$
(B)	Introduction of inert gas at constant pressure will decrease the concentration of reactants	(Q)	$PCl_5\,(g) \rightleftharpoons PCl_3\,(g) + Cl_2\,(g)$
(C)	K_P^o is dimensionless	(R)	$2NO_2\,(g) \rightleftharpoons N_2O_4\,(g)$
(D)	Temperature increase will shift the reaction on product side.	(S)	$NH_3\,(g) + HI\,(g) \rightleftharpoons NH_4I\,(s)$

CE0103

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22.

Column-I (Reactions)	**Column-II (Favourable conditions)**
(A) Oxidation of nitrogen $N_2(g) + O_2(g) + 180.5\ kJ \rightleftharpoons 2NO(g)$	(p) Addition of inert gas at constant pressure
(B) Dissociation of $N_2O_4(g)$ $N_2O_4(g) + 57.2\ kJ \rightleftharpoons 2NO_2(g)$	(q) Decrease in pressure
(C) Oxidation of $NH_3(g)$ $4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g)+6H_2O(g) + 905.6\ kJ$	(r) Decrease in temperature
(D) Formation of $NO_2(g)$ $NO(g) + O_3(g) \rightleftharpoons NO_2(g) + O_2(g) + 200\ kJ$	(s) Increase in temperature

CE0104

MATCHING LIST TYPE :

Answer Q.No. 23, 24 and 25 by appropriately matching the information given in the three columns of the following table.

Column-I (Reaction)	**Column-II (If α is negligiable w.r.t. 1)**	**Column-III**
(A) $2X(g) \rightleftharpoons Y(g) + Z(g),\ \Delta H = -ve$	(P) $\alpha = 2 \times \sqrt{K_p}$	(I) K increase with increase in temperature
(B) $X(g) \rightleftharpoons Y(g) + Z(g),\ \Delta H = +ve$	(Q) $\alpha = 3 \times \sqrt{K_p.P}$	(II) K decrease with increase in temperature
(C) $3X(g) \rightleftharpoons Y(g) + Z(g),\ \Delta H = +ve$	(R) $\alpha = \left(\frac{2Kp}{P}\right)^{1/3}$	(III) Pressure has no effect
(D) $2X(g) \rightleftharpoons Y(g) + 2Z(g),\ \Delta H = +ve$	(S) $\alpha = \sqrt{K_p / P}$	(IV) Addition of inert gas at constant pressure shift equilibrium in forward direction

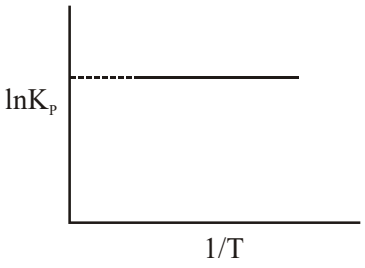
23. Which of the following option is correctly matched ?

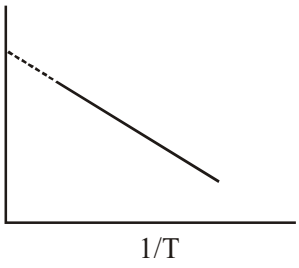
(A) A-Q-III (B) B-S-II (C) C-Q-II (D) D-R-IV

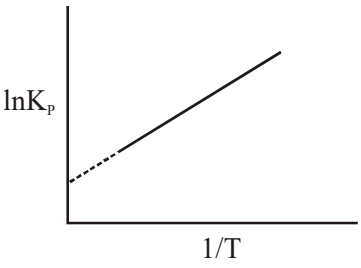
CE0105

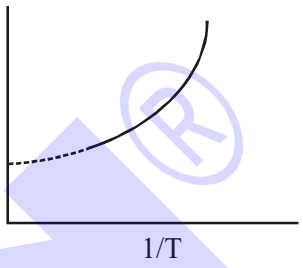
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24. Correct representation of reaction in option (A) is represented by graph

(A) 

(B) 

(C) 

(D) 

CE0106

25. For the reaction of option (B) if degree of dissociation varies inversily as square root of pressure of the system. Suppose at constant temperature volume is increased 16 times of its initial volume the degree of dissociation (α) for this reaction will become :

(A) 4 times (B) 2 times (C) 1/4 times (D) 1/2 times

CE0107

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\14-Ideal-Real-Atomic-Chemical Eq\Eng\14. Che. Eq\02. Ex. p65

EXERCISE # J-MAIN

1. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, K_1 (1) **[JEE-MAINS(online)-13]**

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, K_2 (2)

$H_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons H_2O(g)$, K_3 (3)

The equation for the equilibrium constant of the reaction

$2\ NH_3(g) + \frac{5}{2}O_2(g) \rightleftharpoons 2NO(g) + 3H_2O(g)$, (K_4)

in terms of K_1, K_2 and K_3 is :

(1) $\frac{K_1K_3^2}{K_2}$ (2) $\frac{K_2K_3^3}{K_1}$ (3) $\frac{K_1K_2}{K_3}$ (4) $K_1\ K_2\ K_3$

CE0114

2. In reaction $A + 2B \rightleftharpoons 2C + D$, initial concentration of B was 1.5 times of concentration of A, but at equilibrium the concentrations of A and B became equal. The equilibrium constant for the reaction is :

[JEE-MAINS(online)-13]

(1) 4 (2) 6 (3) 12 (4) 8

CE0115

3. For the decomposition of the compound, represented as **[JEE-MAINS(online)-14]**

$NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

the $K_P = 2.9 \times 10^{-5}\ atm^3$.

If the reaction is started with 1 mol of the compound, the total pressure at equilibrium would be

(1) 38.8×10^{-2} atm (2) 1.94×10^{-2} atm

(3) 5.82×10^{-2} atm (4) 7.66×10^{-2} atm

CE0116

4. For the reaction $SO_{2(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons SO_{3(g)}$, if $K_p = K_C\ (RT)^x$ where the symbols have usual meaning then the value of x is : (assuming ideality) **[JEE-MAINS-14]**

(1) $\frac{1}{2}$ (2) 1 (3) –1 (4) $-\frac{1}{2}$

CE0117

5. The equilibrium constants at 298 K for a reaction $A + B \rightleftharpoons C + D$ is 100. If the initial concentration of all the four species were 1 M each, then equilibrium concentration of D (in mol L^{-1}) will be :

[JEE-MAINS-16]

(1) 1.182 (2) 0.182 (3) 0.818 (4) 1.818

CE0118

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\4-Ideal-Reas-Atomic-Chemical Eq\Eng\4.Che. Eq\02. Ex..p65

6. The following reaction occurs in the Blast Furnace where iron ore is reduced to iron metal:

$Fe_2O_3(s) + 3CO(g) \rightleftharpoons 2\ Fe(l) + 3CO_2(g)$ **[JEE-MAINS(online)-17]**

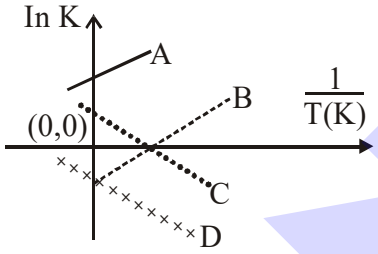
Using the Le Chatelier's principle, predict which one of the following will **not** disturb the equilibrium?

(1) Removal of CO_2 (2) Addition of Fe_2O_3

(3) Addition of CO_2 (4) Removal of CO

CE0119

7. Which of the following lines correctly show the temperature dependence of equilibrium constant, K, for an exothermic reaction ? **[JEE-MAINS(offline)-18]**



(1) B and C (2) C and D (3) A and D (4) A and B

CE0120

8. In which of the following reactions, an increase in the volume of the container will favour the formation of products ? **[JEE-MAINS(online)-18]**

(1) $2NO_2(g) \rightleftharpoons 2NO(g)+O_2(g)$

(2) $3O_2(g) \rightleftharpoons 2O_3(g)$

(3) $H_2(g)+I_2(g) \rightleftharpoons 2HI(g)$

(4) $4NH_3(g)+5O_2(g) \rightleftharpoons 4NO(g)+6H_2O(l)$

CE0121

9. At a certain temperature in a 5 L vessel, 2 moles of carbon monoxide and 3 moles of chlorine were allowed to reach equilibrium according to the reaction,

$CO + Cl_2 \rightleftharpoons CO\ Cl_2$

At equilibrium, if one mole of CO is present then equilibrium constant (K_c) for the reaction is :-

(1) 4 (2) 3 (3) 2 (4) 2.5 **[JEE-MAINS(online)-18]**

CE0122

10. The gas phase reaction $2NO_2(g) \rightarrow N_2O_4(g)$ is an exothermic reaction. The decomposition of N_2O_4, in equilibrium mixture of $NO_2(g)$ and $N_2O_4(g)$, can be increased by :-

(1) Addition of an inert gas at constant volume. **[JEE-MAINS(online)-18]**

(2) Increasing the pressure.

(3) Lowering the temperature.

(4) Addition of an inert gas at constant pressure.

CE0124

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11. For the reaction, **[JEE-MAINS(online)-19]**

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$,

$\Delta H = -57.2 kJ\ mol^{-1}$ and

$K_c = 1.7 \times 10^{16}$.

Which of the following statement is INCORRECT?

(1) The equilibrium constant is large suggestive of reaction going to completion and so no catalyst is required.

(2) The equilibrium will shift in forward direction as the pressure increase.

(3) The equilibrium constant decreases as the temperature increases.

(4) The addition of inert gas at constant volume will not affect the equilibrium constant.

CE0126

12. Two solids dissociate as follows **[JEE-MAINS(online)-19]**

$A(s) \rightleftharpoons B(g) + C(g)\ ;\ K_{p_1} = x\ atm^2$

$D(s) \rightleftharpoons C(g) + E(g)\ ;\ K_{p_2} = y\ atm^2$

The total pressure when both the solids dissociate simultaneously is :-

(1) $(x + y)$ atm (2) $x^2 + y^2$ atm (3) $2\left(\sqrt{x+y}\right)$ atm (4) $\sqrt{x+y}$ atm

CE0128

13. In a chemical reaction, $A + 2B \overset{K}{\rightleftharpoons} 2C + D$, the initial concentration of B was 1.5 times of the concentration of A, but the equilibrium concentrations of A and B were found to be equal. The equilibrium constant(K) for the aforesaid chemical reaction is : **[JEE-MAINS(online)-19]**

(1) 16 (2) 4 (3) 1 (4) $\frac{1}{4}$

CE0129

14. Consider the reaction, **[JEE-MAINS(online)-19]**

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

The equilibrium constant of the above reaction is K_P. If pure ammonia is left to dissociate, the partial pressure of ammonia at equilibrium is given by (Assume that $P_{NH_3} << P_{total}$ at equilibrium)

(1) $\frac{3^{\frac{3}{2}}\ K_P^{\frac{1}{2}}\ P^2}{4}$ (2) $\frac{3^{\frac{3}{2}}\ K_P^{\frac{1}{2}}\ P^2}{16}$ (3) $\frac{K_P^{\frac{1}{2}}\ P^2}{16}$ (4) $\frac{K_P^{\frac{1}{2}}\ P^2}{4}$

CE0130

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\4-Ideal-Reas-Atomic-Chemical Eq\Eng\4. Che. Eq\02. Ex. p65

15. The value of K_p/K_C for the following reactions at 300K are, respectively :
(At 300K, RT = 24.62 $dm^3atm\ mol^{-1}$)

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ **[JEE-MAINS(online)-19]**

(1) 1, 24.62 $dm^3atm\ mol^{-1}$, 606.0 $dm^6atm^2mol^{-2}$

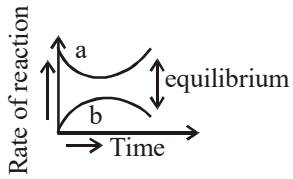
(2) 1, 4.1×10^{-2} $dm^{-3}atm^{-1}\ mol^{-1}$, 606.0 $dm^6\ atm^2\ mol^{-2}$

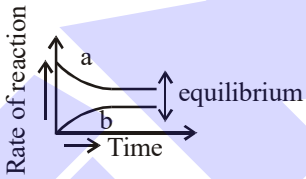
(3) 606.0 $dm^6atm^2mol^{-2}$, 1.65×10^{-3} $dm^3atm^{-2}\ mol^{-1}$

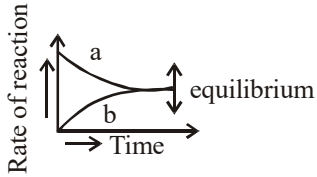
(4) 1, 24.62 $dm^3atm\ mol^{-1}$, 1.65×10^{-3} $dm^{-6}atm^{-2}\ mol^2$

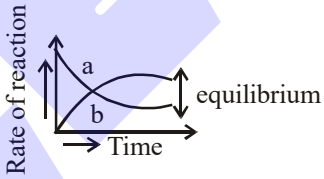
CE0131

16. For the equilibrium $A \rightleftharpoons B$, the variation of the rate of the forward (a) and reverse (b) reaction with time is given by **[JEE-MAINS(online)-2020]**

(1) 

(2) 

(3) 

(4) 

CE0151

17. Consider the following reaction **[JEE-MAINS(online)-2020]**

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)\ ;\ \Delta H^0 = +58\ kJ$$

For each of the following cases (a, b), the direction in which the equilibrium shifts is:

(a) Temperature is decreased

(b) Pressure is increased by adding N_2 at constant T

(1) (a) towards reactant, (b) no change (2) (a) towards product, (b) towards reactant

(3) (a) towards product, (b) no change (4) (a) towards reactant, (b) towards product

CE0152

18. The value of K_C is 64 at 800 K for the reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
The value of K_C for the following reaction is : **[JEE-MAINS(online)-2020]**

$$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$$

(1) $\frac{1}{4}$ (2) $\frac{1}{8}$ (3) 8 (4) $\frac{1}{64}$

CE0153

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19. At 1990 K and 1 atm pressure, there are equal number of Cl_2 molecules and Cl atoms in the reaction mixture. The value K_P for the reaction $Cl_{2(g)} \rightleftharpoons 2Cl_{(g)}$ under the above conditions is $x \times 10^{-1}$. The value of x is _____. (Rounded of to the nearest integer)

[JEE-MAINS(online)-2021]

CE0154

20. A homogeneous ideal gaseous reaction $AB_{2(g)} \rightleftharpoons A_{(g)} + 2B_{(g)}$ is carried out in a 25 litre flask at 27°C. The initial amount of AB_2 was 1 mole and the equilibrium pressure was 1.9 atm. The value of K_P is $x \times 10^{-2}$. The value of x is_____.(Integer answer) **[JEE-MAINS(online)-2021]**

CE0155

21. Consider the reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$. The temperature at which K_C = 20.4 and K_P = 600.1, is_____K. (Round off to the Nearest Integer).

[Assume all gases are ideal and R = 0.0831 L bar K^{-1} mol^{-1}] **[JEE-MAINS(online)-2021]**

CE0156

22. For the reaction, $A + B \rightleftharpoons 2C$

the value of equilibrium constant is 100 at 298 K. If the initial concentration of all the three species is 1 M each, then the equilibrium concentration of C is $x \times 10^{-1}$ M. The value of x is _____ . (Nearest integer) **[JEE-MAINS(online)-2021]**

CE0157

23. The equilibrium constant for the reaction **[JEE-MAINS(online)-2021]**

$$A(s) \rightleftharpoons M(s) + \frac{1}{2}O_2(g)$$

is $K_p = 4$. At equilibrium, the partial pressure of O_2 is _____ atm. **(Round off to the nearest integer)**

CE0158

24. When 5.1 g of solid NH_4HS is introduced into a two litre evacuated flask at 27°C, 20% of the solid decomposes into gaseous ammonia and hydrogen sulphide. The K_p for the reaction at 27°C is $x \times 10^{-2}$. The value of x is _____. (Integer answer) **[Given R = 0.082 L atm K^{-1} mol^{-1}]**

[JEE-MAINS(online)-2021]

CE0159

25. For a reaction at equilibrium

$$A(g) \rightleftharpoons B(g) + \frac{1}{2}C(g)$$

the relation between dissociation constant (K), degree of dissociation ( ) and equilibrium pressure (p) is given by : **[JEE-MAINS(online)-2022]**

(A) $K = \dfrac{\alpha^{\frac{1}{2}} p^{\frac{3}{2}}}{\left(1+\frac{3}{2}\alpha\right)^{\frac{1}{2}}(1-\alpha)}$ (B) $K = \dfrac{\alpha^{\frac{3}{2}} p^{\frac{1}{2}}}{(2+\alpha)^{\frac{1}{2}}(1-\alpha)}$ (C) $K = \dfrac{(\alpha p)^{\frac{3}{2}}}{\left(1+\frac{3}{2}\alpha\right)^{\frac{1}{2}}(1-\alpha)}$ (D) $K = \dfrac{(\alpha p)^{\frac{3}{2}}}{(1+\alpha)(1-\alpha)^{\frac{1}{2}}}$

CE0160

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26. PCl_5 dissociates as

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

5 moles of PCl_5 are placed in a 200 litre vessel which contains 2 moles of N_2 and is maintained at 600 K. The equilibrium pressure is 2.46 atm. The equilibrium constant K_p for the dissociation of PCl_5 is _____ $\times 10^{-3}$. (nearest integer)

(Given: R = 0.082 L atm K^{-1} mol^{-1} : Assume ideal gas behaviour)

[JEE-MAINS(online)-2022]

CE0161

27. $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$

In an experiment, 2.0 moles of NOCl was placed in a one-litre flask and the concentration of NO after equilibrium established, was found to be 0.4 mol/L. The equilibrium constant at 30°C is _____ $\times 10^{-4}$. **[JEE-MAINS(online)-2022]**

CE0162

28. 4.0 moles of argon and 5.0 moles of PCI_5 are introduced into an evacuated flask of 100 litre capacity at 610 K. The system is allowed to equilibrate. At equilibrium, the total pressure of mixture was found to be 6.0 atm. The K_p for the reaction is **[Given : R = 0.082L atm K^{-1} mol^{-1}]**

[JEE-MAINS(online)-2022]

(A) 2.25 (B) 6.24 (C) 12.13 (D) 15.24

CE0163

29. At 298 K, the equilibrium constant is 2×10^{15} for the reaction :

$$Cu(s) + 2Ag^+(aq) \rightleftharpoons Cu^{2+}(aq) + 2Ag(s)$$

The equilibrium constant for the reaction

$$\frac{1}{2}Cu^{2+}(aq) + Ag(s) \rightleftharpoons \frac{1}{2}Cu(s) + Ag^+(aq)$$

is $x \times 10^{-8}$. The value of x is_____. (Nearest Integer) **[JEE-MAINS(online)-2022]**

CE0164

30. At 600K, 2 mol of NO are mixed with 1 mol of O_2.

$$2NO_{(g)} + O_2(g) \rightleftharpoons 2NO_2(g)$$

The reaction occurring as above comes to equilibrium under a total pressure of 1 atom. Analysis of the system shows that 0.6 mol of oxygen are present at equilibrium. The equilibrium constant for the reaction is _____. (Nearest integer). **[JEE-MAINS(online)-2022]**

CE0165

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EXERCISE # J-ADVANCED

1. The thermal dissociation equilibrium of $CaCO_3(s)$ is studied under different conditions.

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ **[JEE 2013]**

For this equilibrium, the correct statement(s) is(are)

(A) ΔH is dependent on T

(B) K is independent of the initial amount of $CaCO_3$

(C) K is dependent on the pressure of CO_2 at a given T

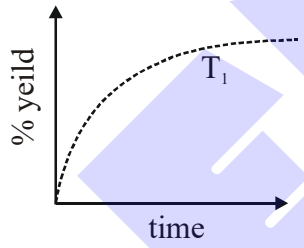
(D) ΔH is independent of the catalyst, if any

CE0134

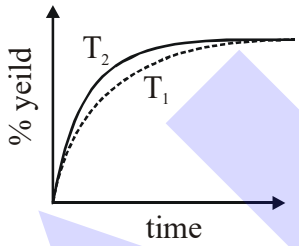
2. The % yield of ammonia as a function of time in the reaction

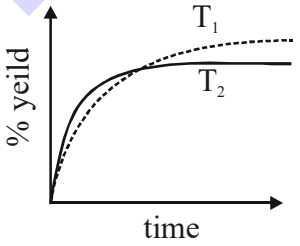
$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g), \quad \Delta H < 0$$

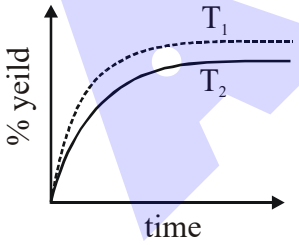
at (P , T_1) is given below -

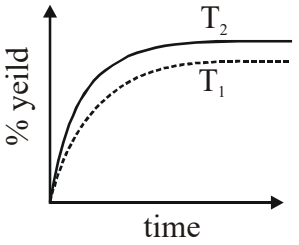


If this reaction is conducted at (P ,T_2), with $T_2 > T_1$, the % yield of ammonia as a function of time is represented by - **[JEE 2015]**

(A)


(B)


(C)


(D)


CE0166

3. For the following reaction, the equilibrium constant K_c at 298 K is 1.6×10^{17}.

$Fe^{2+}(aq) + S^{2-}(aq) \rightleftharpoons FeS(s)$

When equal volumes of 0.06 M $Fe^{2+}(aq)$ and 0.2 M $S^{2-}(aq)$ solutions are mixed, the equilibrium concentration of $Fe^{2+}(aq)$ is found to be $\mathbf{Y} \times 10^{-17}$ **M**. The value of Y is (upto two decimal places.) **[JEE 2019]**

CE0135

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ANSWER KEY

EXERCISE # S-I

1. **(a) negligible (b) almost complete**

2. **(a)** $K_P = \dfrac{(P_{SO_3})}{(P_{SO_2})(P_{O_2})^{\frac{1}{2}}}$; $K_C = \dfrac{[SO_3]}{[SO_2][O_2]^{\frac{1}{2}}}$

(b) $K_C = [H_2O]$

$K_P = P_{H_2O}$

(c) $K_C = [NH_3]^2 [CO_2]$

$K_P = \left(P_{NH_3}\right)^2 P_{CO_2}$

(d) $K_C = \dfrac{[CH_3COOC_2H_5][H_2O]}{[CH_3COOH][C_2H_5OH]}$ **[here H_2O is not in excess]**

K_P is not defined for liquid phase.

(e) $K_C = \dfrac{[NH_4^+][OH^-]}{[NH_3]}$ **[here $H_2O(\ell)$ is in excess (solvent) so its concentration is 1]**

(f) $K_{PC} = \dfrac{P_{H_2} \times [Zn^{2+}]}{[H^+]^2}$

3. **(a)** **Here, $\Delta n_g = 1+ 1 - 1 = 1$**

$K_p = K_c (RT)^1 = K_c RT$

(b) **Here, $\Delta n_g = 2 - (3+1) = -2$**

$K_p = K_c (RT)^{-2}$

(c) **Here, $\Delta n_g = 2 - (1+1) = 0$**

$K_p = K_c (RT)^0 = K_c$

(d) **Here, $\Delta n_g = 2 + 1 - 2 = 1$**

$K_p = K_c (RT)^1 = K_c RT$

(e) **Here, $\Delta n_g = 2 - (2 + 1) = -1$**

$K_p = K_c (RT)^{-1} = K_c / RT$

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\4-Iblock-Rass-Atomic-Chemical Eq\Eng\4. Che. Eq\02. Ex. p65

4. XO_2

5. $Q_c > K_c$ so the reaction will proceed in the reverse direction.

6. (a) $Q_C = \frac{[CH_3COOC_2H_5][H_2O]}{[CH_3COOH][C_2H_5OH]}$ (b) 3.48 (c) No

7. 2.00 8. 0.90 9. 100.00 10. 0.24, 0.01

11. No, 6.40×10^{-31} M 12. 0.44 13. 1.20

14. 0.33 15. (a) 0.10, (b) 0.40 16. 1.00

17. 4.00 18. $a = \sqrt{\frac{K_p}{4P + K_p}}$ 19. 16.00 20. 2.00

21. 0.50 22. 0.05 23. 0.09 or 0.10 24. 50.00

25. 0.01 26. 15.00 27. 4.00 28. 1.80

29. 22.40 30. 5.00 31. 0.82

32. (a) $K = [CH_3OH]/[H_2]^2[CO]$,

(b) 1. $[H_2]$ increase, [CO] decrease, $[CH_3OH]$ increase ; 2. $[H_2]$ increase, [CO] decrease, $[CH_3OH]$ decrease ; 3. $[H_2]$ increase, [CO] increase, $[CH_3OH]$ increase ; 4. $[H_2]$ increase, [CO] increase, $[CH_3OH]$ increase ; 5. $[H_2]$ increase, [CO] increase, $[CH_3OH]$ decrease ; 6. no change

33. 6

34. 250.00

35. k_r increase more than k_f, this means that E_a (reverse) is greater than E_a (forward). Hence exothermic reaction.

EXERCISE # S-II

1. (2.40) 2. (3.44)

3. (0.11) 4. (0.20)

5. (0.36 or 0.37)

6. (a) 6.66 or 0.67; (b) 375.00; (c) 10.45 or 10.46; (d) 6.25

7. (i) 2; (ii) 1.20 8. 54.00

9. 3.60 10. (a) 400, 900, (b) 4: 9, (c) 72.15

11. (4.00) 12. $(SrCl_2 \cdot 2H_2O)$

13. (10.00) 14. $K_{p_1} = \frac{1}{20P_0^2}$; $K_{p_2} = \frac{3}{20P_0^2}$

15. (a) 80 ; (b) 0.15 ; (c) 26.08 or 26.09

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\4-Ideal-Reas-Atomic-Chemical Eq\Eng\4.Che. Eq\02. Ex..p65

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	C	D	B	C	C	A	C	A	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	C	C	A	B	B	A	D	B	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	B	A	C	C	C	B	C	D
Que.	31	32	33	34	35					
Ans.	A	A	C	A	D					

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C,D	C,D	A,B,C,D	C,D	A,C,D	A,B,C	C,D	A,C	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	A	A	B	C	B	A	D	A	A
Que.	21				22				23	24
Ans.	A-P,R,S	B-P,Q,R	C-P,Q,R,S	D-Q	A-s	B-p,q,s	C-p,q,r	D-r	D	C
Que.	25									
Ans.	A									

EXERCISE # J-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	1	3	4	4	2	4	1	4	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	3	2	2	4	3	1	2	5	73
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	354	25	16	6	B	1107	125	A	2	2

EXERCISE # J-ADVANCED

Que.	1	2	3
Ans.	A,B,D	B	8.92 or 8.93

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Important Notes

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\4-Ideal-Real-Atomic-Chemical Eq\Eng\4.Che. Eq\02. Ex. p65